

MCGREGOR, TX, USA

WMO: 722563

Lat: 31.485N

Lon: 97.316W

Elev: 180

StdP: 99.18

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53952

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	-2.4	-12.2	1.3	-0.9	-9.6	1.7	1.6	11.9	15.3	10.9	13.4	3.8	0	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	38.7	23.7	37.6	23.8	36.5	23.8	25.8	33.4	25.5	33.2	25.1	32.7	4.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.4	28.1	23.6	18.9	27.9	23.0	18.1	27.7	81.2	33.6	79.1	33.3	77.3	32.8	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.2	8.3	-7.9	40.5	2.4	1.6	-9.6	41.6	-11.0	42.6	-12.4	43.5	-14.1	44.6		
			-8.9	26.7	2.1	0.7	-10.4	27.2	-11.6	27.7	-12.8	28.1	-14.3	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	8.9	11.1	14.7	19.3	23.7	27.7	29.5	29.7	25.9	20.2	14.4	9.6
	DBStd	8.57	5.60	5.78	5.24	4.11	3.51	2.35	1.93	2.14	3.40	4.42	5.16	5.33
	HDD10.0	251	88	52	19	1	0	0	0	0	0	1	18	73
	HDD18.3	1129	295	211	135	38	5	0	0	0	2	33	136	274
	CDD10.0	3752	54	81	164	279	424	529	604	609	478	316	151	61
	CDD18.3	1590	3	7	22	66	171	279	346	351	230	91	19	4
	CDH23.3	18386	16	56	143	544	1630	3269	4506	4654	2554	861	130	23
	CDH26.7	9499	1	11	21	147	670	1703	2599	2716	1293	313	23	2
Wind	WSAvg	4.0	4.1	4.4	4.6	4.7	4.3	4.0	3.7	3.5	3.3	3.8	3.9	3.9
Precipitation	PrecAvg	931	67	70	93	74	112	91	51	65	80	104	80	78
	PrecMax	1227	138	173	249	170	325	268	141	224	238	280	260	220
	PrecMin	639	15	4	36	16	18	19	0	1	12	15	12	18
	PrecStd	170	40	47	48	43	67	60	39	71	59	69	63	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.6	27.9	28.8	32.7	36.1	38.8	40.1	41.1	39.0	33.7	29.1	26.2
		MCWB	16.3	17.3	19.1	21.2	23.0	23.6	23.6	23.8	22.9	22.4	20.9	18.1
	2%	DB	22.6	24.9	27.0	30.1	33.7	37.2	38.3	38.8	36.3	32.2	27.0	22.9
		MCWB	16.0	16.7	18.7	20.4	23.2	23.8	23.6	24.0	23.3	21.7	19.4	17.3
	5%	DB	20.9	22.5	24.9	27.9	32.3	35.9	37.4	37.7	34.7	30.2	24.1	21.2
		MCWB	15.5	16.3	17.8	20.1	23.0	24.0	23.8	23.9	23.2	20.9	18.1	16.4
	10%	DB	18.0	20.9	22.7	26.8	30.9	33.9	36.2	36.9	32.9	28.0	22.6	18.8
		MCWB	14.0	16.2	17.0	19.3	22.6	23.9	23.9	23.9	22.7	19.6	17.7	15.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	20.5	21.5	24.0	25.8	26.5	26.2	26.7	25.9	24.9	22.7	20.5
		MCDB	22.4	23.8	26.1	29.1	31.0	33.2	34.1	34.2	33.1	30.3	26.9	23.4
	2%	WB	18.1	19.2	20.3	22.7	24.9	25.7	25.7	25.8	25.2	23.8	21.1	19.0
		MCDB	20.7	22.4	24.3	27.7	30.6	32.9	33.7	33.9	32.0	28.8	24.6	22.0
	5%	WB	16.7	18.0	19.4	21.7	24.1	25.2	25.3	25.4	24.6	22.8	19.8	17.4
		MCDB	19.2	21.0	23.1	26.5	29.7	32.5	33.4	33.6	31.0	27.3	23.2	19.9
	10%	WB	14.7	16.5	18.4	20.6	23.4	24.7	24.9	24.9	24.0	21.7	18.6	15.3
		MCDB	18.1	20.2	21.8	24.9	28.9	31.6	32.8	33.0	30.0	26.1	22.1	18.3
Mean Daily Temperature Range	5% DB	MDBR	11.8	11.8	11.8	12.4	11.5	11.8	12.6	12.7	12.5	12.7	11.6	11.5
		MCDBR	14.6	14.3	13.1	13.4	13.2	13.8	14.2	14.4	14.2	13.9	13.1	14.0
		MCWBR	9.5	9.1	7.7	7.1	5.3	4.2	3.9	4.2	5.0	6.7	7.8	9.5
	5% WB	MCDBR	11.2	11.6	10.8	10.5	10.9	11.4	12.2	12.4	11.6	10.8	10.5	11.4
		MCWBR	8.8	8.7	7.3	6.4	5.3	4.4	3.9	3.9	4.5	5.9	7.4	9.1
Clear-Sky Solar Irradiance	taub	0.316	0.335	0.358	0.399	0.429	0.431	0.443	0.436	0.401	0.370	0.347	0.323	
	taud	2.509	2.463	2.413	2.322	2.296	2.325	2.326	2.332	2.375	2.447	2.456	2.493	
	Ebn at Noon	913	925	924	893	864	857	845	849	868	874	866	883	
	Edh at Noon	87	100	113	129	133	129	128	126	116	101	91	84	
All-Sky Solar Radiation	RadAvg	2.96	3.43	4.49	5.43	5.78	6.58	6.72	6.22	5.21	4.16	3.12	2.60	
	RadStd	0.31	0.58	0.39	0.49	0.51	0.53	0.48	0.37	0.43	0.62	0.38	0.36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page